

Workshop *Part-Whole Structure and its Reflection in Natural Language*

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Notions of a Whole in Semantics

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The aim of the talk

Give a very general, informal overview of notions of a whole that have been used or that are important for the semantics of natural language.

1. Notions of a whole in extensional: the notions of a sum and of an atom
 2. Notions of a whole based on structural relations among parts: the notions of an integrated whole and of boundedness
 3. Notions of a whole as prior to the parts
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1. Notions of a whole in extensional mereology

1.1. Assumptions of extensional mereology (Link 1983, Cotnoir/Varzi 2021)

Unstructured parts and wholes

- Transitive part relation
- Entities that share the same parts are identical / uniqueness of sums

Sum formation or fusion *sum*

(1) For a nonempty set P , $\text{sum}(x, P) =_{\text{def}} \forall y (y \text{ O } x \leftrightarrow \exists z (y \text{ O } z \ \& \ P(z)))$

A sum of a set P is a thing such that everything which overlaps with it also overlaps with something in P , and vice versa.

(2) Uniqueness of sums: $\forall P (P \neq \emptyset \rightarrow \exists! z \text{ sum}(z, P))$

1.2. Sum as notion of a whole in extensional mereology

Uses of sums in semantics

1) Definite NPs:

- (3) a. The children: $\iota x[\text{sum}(x, \text{children})]$
 b. The water: $\iota x[\text{sum}(x, [\text{water}])]$
- 2) Non-Boolean conjunction: conjunction interpreted by sum formation ($\oplus$, sum)
- (4) a. John and Mary embraced.
 b. The men and the women gathered.
 c. The oil and the vinegar were mixed.
 d. The students were French and German.
- (5) a. $E(j \oplus m)$
 b. $G(\iota x \text{sum}(x, M \cup W))$
 c. $M(\iota x \text{sum}(x, O \cup V))$

Issues with sum as a notion of a whole

Sum formation does not apply to entities to yield individuals:

- (6) a. the hot dog = the sausage and the bun?
 b. The sausage and bun taste similar / ?? costs 4 dollar.
 c. The hot dog costs 4 dollar / ?? tastes similar.

Note, however, that non-Boolean conjunctions of adjective conjunctions are applicable to single individuals:

- (7) The Italian flag is red, green and white.

Non-Boolean adjective conjunction is domain-neutral:

- (8) a. the blue and while pillow
 b. the blue and white pillows
 c. the blue and white laundry

States: like individuals, complex states cannot be formed through sum formation

- (9) a. John's being calm and nervous $\neq$ John's being calm and John's being nervous
 b. John's being calm and John's being nervous do not go together.
 c. ??? John's being calm and nervous does not go together.
- (10) a. John's hope to win and to become rich makes sense.
 b. ? John's hope to win and his hope to become rich make sense.

Conclusion

Sum formation strictly yield entities in the domains of pluralities and quantities, not individuals (denotable by singular count NPs).

Special case: semantics of non-Boolean adjective conjunction

2.3. Atom as a notion of a whole in extensional mereology

The formal semantics of plurals and mass nouns based on extensional mereology requires distinguishing three domains each with its own part relation:

The domain of individuals I

The domain of pluralities P

The domain of quantities (or portions or stuff) M

$(P, \oplus_i)$ is a join semilattice i.e., $(P, <_i)$ is an extensional mereology.

$(M, \oplus_m)$ is a join semilattice, i.e., $(M, <_m)$ is an extensional mereology.

(11) For a plural noun extension N_{pl} , $N_{pl} = \oplus N$

Necessity to distinguish two part relations $<_i$ and $<_m$, in addition to the part relation $<$ applying to parts of individuals.

A part of something described as a ‘sum’ is neither a part in the sense of $<_i$ nor in the sense of $<_m$, but only in the sense of $<$.

A part of a plurality is never a part in the sense of $<$ or $<_m$.

Individuals are atoms with respect to $<_i$, but not generally with respect to $<$, and

$<_m$ does not apply to pluralities or individuals (except in Link 1983).

The role of the notion of an atom in the semantics of singular count, plural, mass nouns

Two notions of an atom:

(12) For a singular count noun extension N:

for all x, if $N(x)$, then x is an atom with respect to $<_i$ or with respect $<_i$ and N.

(13) a. $\text{atom}(x) =_{\text{def}} \neg \exists y (y <_i x \ \& \ x \neq y)$

b. For a set N, $\text{atom}(x, N) =_{\text{def}} \neg \exists y (y \in N \ \& \ y < x \ \& \ x \neq y)$

Potential problem: sequence-type nouns

Continuous parts of sequences are again sequences; similarly for sums, entities, quantities, fences, walls, Russian dolls

These do not actually pose a problem for (13a), but only for (13b).

A part of a sequence is not a part in the sense of $<_i$, but only in the sense of $<$, since a sequence is not a proper plurality. A sequence and its parts are atoms – with respect to $<_i$!

(Potential) motivations for a single part relation (Moltmann 1997)

1. Domain-neutral special quantifiers:

(14) a. What is in the bag? An apple, some bread, some coins.

b. John ate something, bread, an apple, or some peas.

2. Predicates with domain-independent part structure-related content

(15) a. the frequent rain

b. the frequent rainfalls

Lexical condition of temporal separation of parts, in the generalized sense of ‘part’

Worries concerning the notion of an atom

1. Does the notion of atom really capture the mass-count distinction, by explaining countability, the notion of being a single object? (What is wrong with counting parts of Russian dolls etc...?)

2. Extensional mereology with its distinction into plural-specific and mass-specific domains involves strictly language-dependent part structures, dependent on the use of plural or mass categories.

2. Structured wholes

2.1. Problems for extensionality

Different structural or functional wholes with the same parts:

Collection of lines – triangle

Wood – chair

Wood – pile of wood

Water – puddle of water

Plurality of people – committee

Committee 1 – committee 2

Structural wholes / integrated wholes

Wholes as built from parts and structural relations among them.

2.2. Notions of structure

Spatial relations among parts (for material objects)

Temporal relations among parts (for events)

Function of parts with respect to the function of the whole (e.g., committee)

A simple notion of an integrated whole:

A maximally self-connectedness entity (R-integrated whole) (Simons 1987, Moltmann 1997)

(17) R-integrated whole

For a symmetric, reflexive non-formal relation R, for the transitive closure R_{trans} of R:

X is an R-integrated whole if for any y, z, $y <_R x$, $y <_R x$, $y R_{\text{trans}} z$ and for no y, z, $y <_R x$,

$\neg z <_R x$, $y R_{\text{trans}} z$.

(18) Transitive closure of a relation R

For a relation R, $x R_{\text{trans}} y$ iff for entities $x_1, \dots, x_n$, $x R x_1$ & $x_1 R x_2$... & $x_n R y$

Examples of relations R

Spatial contiguity, temporal contiguity, kinship relation

Being part of the same class, family, committee

Special case of an R-integrated whole:

(19) FF-integrated whole

Relation FF defined in terms of a property F: $x FF y$ iff $F(x)$ and $F(y)$.

Examples

(20) a. the blue things. $F =$ being blue.

b. The children in the garden $F =$ child in the garden

Some linguistic applications

The readings of part-related predicates *compare* and *distinguish* (Moltmann 1997)

FF-integrated wholes:

(21) a. John cannot distinguish the boys and (the) girls.

b. Mary compared the blue and green balls.

R-integrated wholes

- (22) a. ? Mary compared the furniture.
 b. Mary compared the furniture in the two rooms.
 R: 'be in the same room as'

The meaning of *times* involves maximally continuous processes or states (Moltmann 1997)

- (23) a. John came several times today.
 b. John slept several times today.

General questions

Are integrated wholes always objects?

No:

Pluralities that are integrated wholes (e.g., 'the students at the different universities')

Quantities that are integrated wholes (e.g., 'the content of the two boxes')

Application to the mass-count distinction

- (24) a. For a singular count noun N, if N holds of an entity x, then x is an (essential) integrated whole.
 b. For a singular count noun N and a reference situation, if N holds of an entity x, then x is an integrated whole in s (Moltmann 1997)
 c. For a singular count noun N, N conveys a property of integrated wholes.

Langacker (1987):

- (25) If a singular count noun holds of an entity x, then x is a bounded region in some dimension. (Avoids the problem of sequence-type nouns.)

The mass-count distinction for parts:

Part of vs. *a part of* (e.g., Moltmann 1998)

- (26) a. John and Mary are part of the class.
 b. ?? John and Mary are a part of the class.
 (27) a. Sugar is part of lemonade.
 b. ??? Sugar is a part of the lemonade.
 (28) a. Joe ate a part of the cake.
 b. Joe ate part of the cake.

Notions of integrated wholes play also roles in choices of nominalizations for verbs or adjective and conversions of mass nouns to count nouns and conversely.

General worries

1. Is integrity necessary for something to count as a single thing, to be 'one'? (Moltmann, to appear)

(29) a. the quantities of water in this glass

b. the amount of water the body consists in

c. the portions of rice on this plate

Singular count nouns may convey merely conceived integrity? (Langacker 1987, Moltmann 1997)

2. Even pluralities and quantities may be structured or integrated wholes, without counting as 'one'

(30) a. John cannot distinguish / ??? count the rice and the beans. (counting two)

b. ??? The students are one of the people I presented my work to.

c. The students are among the people I presented to.

3. Notion of a whole as prior to the parts

3.1. Functional parts

Door knobs, doors, handles: cannot be understood without a conceived whole.

3.2. Cases of failure of transitivity of the part relation

Invalid:

(31) John is part of the class

John's leg is part of John.

John's leg is part of the class.

Valid:

(32) The arm is part of the body.

The hand is part of the arm.

The hand is part of the body.

Invalid:

(33) The page is part of the book.

The book is part of the library.

The page part of the library.

Valid:

(34) The page is part of Goethe's written work.

The book is part of Goethe's written work

The page part of Goethe's written work.

Written work as mass nominal allows for transitivity, to some extent, but not for inferences with 'the empty pages of the book', 'the margins of the page', etc.

It is not the integrity of the intermediary entity that may block transitivity; rather the whole determines what its parts are.

Potential ways of dismissing problems for transitivity

1) dismiss functional parts as the only parts

Issue: what is the intuitive basis for the part relation, if not the applicability of *part of*?

2) distinguish different part relations, for different ontological levels:

Part relation for individuals, part relation for pluralities, part relation for stuff

Issue: transitivity problems arise within the levels of individuals, of pluralities, and of quantities as well (e.g., higher-level pluralities)

3.3. Expressions of completion

Expressions of completion, *partial(ly)*, *complete(ly)*, involve reference to a conceived whole.

(35) the complete / partial realization of the plan

Conceived whole? Plans are conceived wholes!

Expressions of completion may make reference to particular kinds of wholes.

Two words for 'complete' in German: *voellig* and *vollstaendig*

(35) a. die vollstaendige Uebersetzung

‘the complete translation’

b. ??? die voellige Uebersetzung

(36) a. die voellige Dunkelheit

‘the complete darkness’

b. ??? die vollstaendige Dunkelheit

Vollstaendig relates to discrete part structure; *voellig* to a homogenous part structure.

3.5. Completion-related absence

Absence of something that should be there in order for something else to be complete.

Completion-related predicates of absence:

lack, be missing, German: *mangeln, fehlen*, French: *manquer*, Italian: *mancare*

Only things that can be ‘had’ can ‘lack’:

(37) a. The house has a door

b. The house lacks a door

(38) a. John has a mother / a car

b. John lacks a mother / a car.

Lack is a modal notion: lack $\neq$ not have:

(39) a. Mary has a ponytail.

b. ?? Mary lacks a ponytail.

(40) a. The house has a balcony.

b. ?? The house lacks a balcony.

When acceptable, (39b) entails:

(41) Mary should have a ponytail.

Application condition for *lack*

Lack relates to a conceptual whole and presupposes that that conceptual whole is not fully manifested or realized.

The conceptual whole need not have objects as its realizations:

The whole may be constituted instead by relations of possession, kinship, friendship.

The whole may also be a plurality:

(23) a. The students were in the hallway, but John was missing.

b. Bill saw the students, but John was missing.

More uses of *lack*

Have and *lack* can also relate an individual to a quality:

(42) a. Joe has wisdom.

b. Joe lacks wisdom.

(43) a. Mary has talent.

b. Mary lacks talent.

(44) a. Mary's lack of understanding was astonishing.

b. Mary's lack of attention to details ruined the project.

Particularized qualities (tropes) are not really parts of objects.

Location-related *lack*:

(45) a. There is a lack of water

b. There is water.

Generalization

Lack involves the notion of an integrated wholes that is itself not tied to single objecthood.

More general conclusion

The notion of an integrated whole is independent of the notion of a single object.

3.4. Predicates of replacement

Replacement can apply only to well-delimited, often functional parts:

(46) Mary replaced the wheel / the table top / the screw.

Replace cannot apply to qualities, surfaces, appearances of objects:

(47) ??? Mary replaced the color / the texture / the weight / the surface / the appearance of the object.

Replacement means taking away a structural or functional part and putting a similar or equivalent object in its place.

Conclusions

1. The notion of a whole available in extensional mereology is clearly insufficient for the semantics of natural language, if not problematic in the first place.
2. The notion of an integrated or structural whole as a whole built from parts and relations among parts plays an important in natural language.

3. In addition, a notion of a whole that is prior to the parts plays an important role in phenomena that so far have received much less attention.

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